Factors associated with player injury in the Midland Premier League 2023/24

MTH6134 Statistical Modelling II — Group Project, Week 3: Introduction to GLMs

1. Introduction

This report investigates which player characteristics are associated with whether a player was injured during the (fictional) 2023/24 Midland Premier League season. The aim is to use a generalised linear model (GLM) to describe how age, playing position, training load, number of previous injuries, body mass index (BMI) and minutes played relate to the probability of injury. In brief, injury was more likely for older players, for players with a higher training load and for players with more previous injuries (Table 3), whereas position, BMI and minutes played were not needed once these three were included. Diagnostic checks found no individual observation that unduly influences the fit, but they indicate that the association between training load and injury may depend on age (Section 4.4).

2. Data

The players dataset (*league_players.csv*) has one row per player for 620 players from all 24 clubs (24–28 players per club) and eight variables (Table 1). There were no missing values or duplicated rows, and every variable lay in a plausible range (for example, no player exceeded the 4,140 minutes available in a 46-match season). Of the 620 players, 115 (18.5%) were recorded as injured.

Choice of response. The natural response in the players data is *injured*, a binary outcome (1 = injured, 0 = not). In the Ashford match data the natural responses are *goals_scored* and *goals_conceded*, which are counts, but there are only 46 matches and these are better suited to Poisson regression later in the module. *minutes_played* is continuous but is probably affected by injury (injured players miss matches), so we treat it as an explanatory variable with caution rather than as a response. We therefore model *injured*, which also has the most observations and several plausible explanatory variables.

Table 1. Variables in the players dataset, their type, role in the model and summary statistics.

Variable	Type	Role	Summary (n = 620)
injured	Binary (0/1)	Response	115 injured (18.5%)
age	Discrete numeric (years)	Explanatory	Mean 25.1, SD 3.9, range 17–36
position	Nominal (4 levels)	Explanatory	DEF 216, MID 206, FWD 136, GK 62
training_load	Ordinal 1–5, used as numeric	Explanatory	Levels 1–5: 20, 133, 239, 158, 70 players
prev_injuries	Count (0–4)	Explanatory	0 to 4 injuries: 295, 171, 94, 39, 21 players
bmi	Continuous	Explanatory	Mean 23.5, SD 1.5, range 19.0–27.9
minutes_played	Continuous	Explanatory	Mean 2540, SD 710, range 105–3960
team	Nominal (24 clubs)	Grouping (Section 4.3)	24–28 players per club

Figure 1 shows the observed proportion of players injured. It rises steadily with the number of previous injuries (10.8% with none, 42.9% with four) and, from age 22–24 onwards, with age (11.8% rising to 30.4% for ages 31–36). It is also higher at training loads 4 and 5 (about 26%) than at load 2 (11.3%).

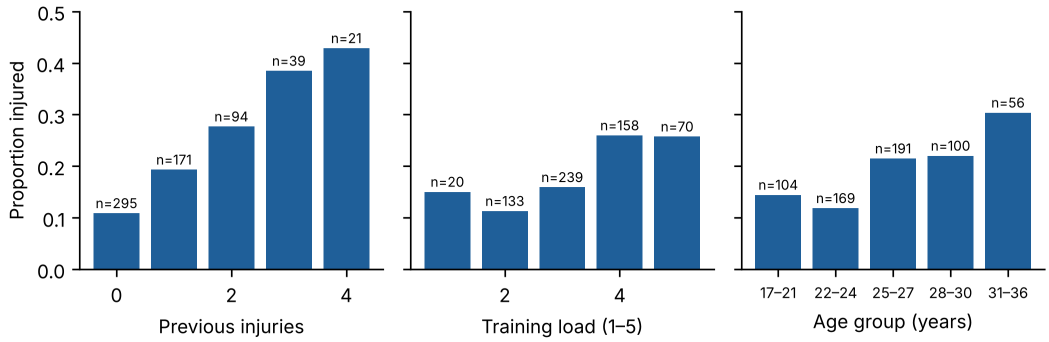


Figure 1. Observed proportion of players injured by number of previous injuries, training load and age group. Bar labels give the number of players in each group.

3. Methods

Because *injured* takes only the values 0 and 1, we model it as $Y_i \sim \text{Bernoulli}(\pi_i)$ with a logit link, $\log\{\pi_i/(1-\pi_i)\} = \beta_0 + \beta_1 x_{i1} + \ldots + \beta_k x_{ik}$. This binomial GLM (logistic regression) keeps fitted probabilities between 0 and 1 and allows the variance $\pi_i(1-\pi_i)$ to depend on the mean, which a normal linear model cannot do for a 0/1 response. The exponential of a coefficient, $\exp(\beta_j)$, is the odds ratio for a one-unit increase in x_j with the other variables held fixed. Position was entered as a factor with defenders (DEF) as the reference level.

We began with all six explanatory variables (the full model), tested each term using likelihood ratio (LR) tests, and compared nested models using an LR test and AIC. Confidence intervals are profile-likelihood intervals. Goodness of fit was assessed with the Hosmer–Lemeshow test, and the final model was checked using binned residuals, standardised deviance residuals, leverage and Cook's distance. All code is in the Appendix.

4. Results

4.1 Model selection

In the full model (AIC 554.22) the coefficient of *minutes_played* was essentially zero (p = 0.96). Table 2 gives LR tests for removing each term from the full model. Age, training load and previous injuries each showed clear evidence of association (all p < 0.01), whereas position (p = 0.51), BMI (p = 0.58) and minutes played (p = 0.96) did not. Removing these last three terms together increased the deviance by only 3.05 on 5 degrees of freedom (LR test p = 0.69) and lowered AIC from 554.22 to 547.27. We therefore selected the model containing age, training load and previous injuries. This is a decision about which terms are needed, not proof that the others are unrelated to injury: for example, goalkeepers had the highest observed injury rate (27.4% against 15.7% for defenders), but this difference was not supported after adjusting for the other variables, and with only 62 goalkeepers the data have limited power to detect it.

Table 2. Likelihood ratio tests for each term in the full model (residual deviance 536.22, AIC 554.22).

Term dropped from full model	df	LR statistic	p-value	AIC without term
age	1	13.75	< 0.001	565.97
training_load	1	6.97	0.008	559.20
prev_injuries	1	30.86	< 0.001	583.09
position	3	2.32	0.508	550.55
bmi	1	0.30	0.581	552.53
minutes_played	1	0.002	0.964	552.23

4.2 Final model and interpretation

The final model is logit(π) = −5.789 + 0.1053 × age + 0.3201 × training_load + 0.5113 × prev_injuries (residual deviance 539.27 on 616 degrees of freedom; AIC 547.27). Table 3 gives the estimates as odds ratios. Holding the other two variables fixed, each additional year of age is associated with 11.1% higher odds of injury (about 1.69 times the odds per five years), each additional point of training load with 37.7% higher odds, and each additional previous injury with 66.7% higher odds. All three confidence intervals exclude 1, in agreement with the LR tests. These are odds ratios, not changes in probability: the fitted probability of injury is 5.6% for a 22-year-old with training load 2 and no previous injuries, and 54.6% for a 30-year-old with training load 4 and three previous injuries. The data are observational, so these are associations rather than causal effects.

Table 3. Final logistic regression model for injury. Confidence intervals are profile-likelihood intervals; p-values are from Wald z-tests.

Term	Estimate (log-odds)	SE	Odds ratio	95% CI for odds ratio	p-value
Intercept	−5.789	0.859	–	–	< 0.001
Age (per year)	0.1053	0.0291	1.111	1.050 to 1.177	< 0.001
Training load (per point)	0.3201	0.1103	1.377	1.111 to 1.714	0.004
Previous injuries (per injury)	0.5113	0.0911	1.667	1.396 to 1.997	< 0.001

4.3 Goodness of fit and diagnostics

Goodness of fit. The residual deviance (539.27 on 616 df) and the Pearson statistic divided by its degrees of freedom (656.0/616 = 1.065) are often compared with their degrees of freedom, but with an ungrouped 0/1 response these comparisons are not valid tests of fit: the deviance depends on the data only through the fitted probabilities and is not approximately χ^2, and overdispersion cannot be assessed in this way. We therefore used the Hosmer–Lemeshow test with 10 groups, which gave $\chi^2 = 14.91$ on 8 df ($p = 0.061$): borderline evidence that the model does not fit perfectly. The discrepancy comes mainly from the lowest-risk group, where 8 injuries were observed against 3.3 expected, and the highest-risk group (34 observed, 28.8 expected). The model's discrimination is moderate (area under the ROC curve 0.70).

Residuals. Plots of deviance or Pearson residuals against fitted values for a binary response form two separate curves (one for each outcome) by construction and so are uninformative. We instead used binned residuals (Figure 2a): 19 of 20 bins lie inside approximate 95% bands, with the small positive residuals at the lowest fitted probabilities matching the Hosmer–Lemeshow result. Twenty-six players had a standardised deviance residual with absolute value above 2 (maximum 2.53, none above 3); all 26 were injured players with low fitted probabilities, i.e. unexpected injuries.

Leverage and influence. Mean leverage is p/n = 0.0065; 54 players exceeded the 2p/n cut-off of 0.0129 (maximum 0.0297). Cook's distance exceeded 4/n = 0.0065 for 51 players, but the maximum was only 0.0255 (Figure 2b), far below 1. Only 8 players were flagged by both rules. These cut-offs are rough screening rules and, with 620 players, flagging about 8–9% is unsurprising; no single observation stands out. Of the 51 Cook's-distance flags, 48 were injured players. Since all values are plausible and these are genuine outcomes rather than apparent errors, we did not remove any observations: doing so would remove the very injuries the model is meant to explain.

Clustering by club. The model treats players as independent, but they come from 24 clubs. Adding club as a factor to the final model gave an LR statistic of 20.88 on 23 df ($p = 0.59$), so there is no evidence that clubs differ in injury risk beyond what the three predictors explain. Models that allow for grouping are considered later in the module.

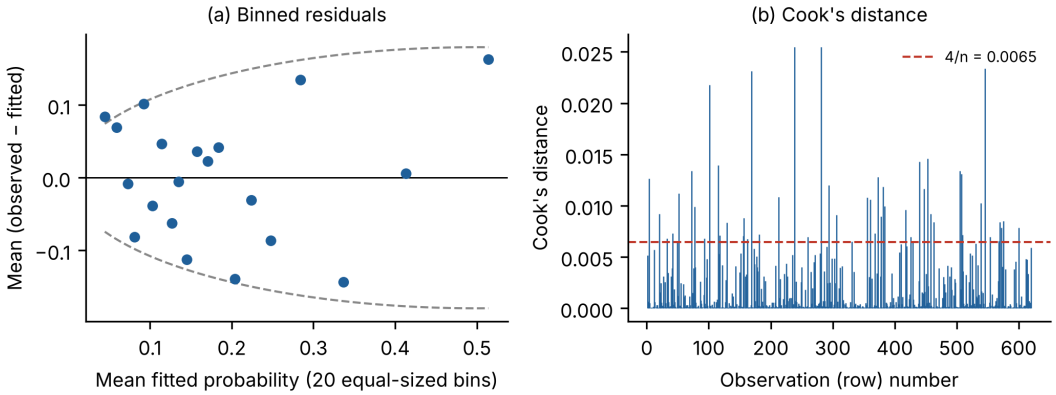


Figure 2. Diagnostics for the final model. (a) Binned residuals (20 bins of 31 players) against mean fitted probability, with approximate 95% bands (dashed). (b) Cook's distance for each player, with the 4/n cut-off (dashed).

4.4 Checking linearity and interactions

Adding quadratic terms for age, previous injuries or training load (p = 0.40, 0.50 and 0.92), or treating training load as a factor (p = 0.24), did not improve the fit, so the linear terms on the logit scale are adequate. Of the three pairwise interactions, age × previous injuries (p = 0.78) and training load × previous injuries (p = 0.12) were not supported, but age × training load was: adding it reduced the deviance by 13.97 on 1 df (p < 0.001) and lowered AIC from 547.27 to 535.30 (interaction coefficient 0.111, SE 0.031). Figure 3 shows the pattern: among players aged 27 or over the proportion injured rises from 10.0% at training load 1–2 to 39.3% at load 4–5, while for players aged 22 or under it does not rise (11.4% to 9.6%). Under the interaction model the odds ratio per point of training load is about 0.89 at age 22, 1.24 at age 25 and 2.17 at age 30, so the single training-load odds ratio in Table 3 is an average over ages.

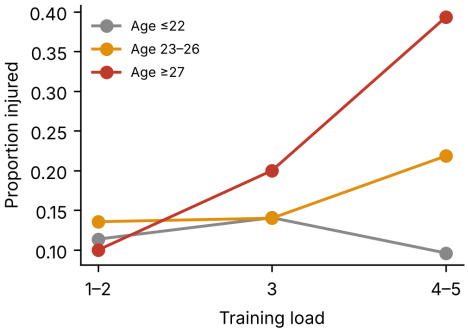


Figure 3. Observed proportion of players injured by training load, in three age groups (groups of 160, 246 and 214 players). The rise with training load is concentrated among older players.

5. Conclusions

In this simulated season, the odds of injury were higher for older players, for players with a higher training load and for players with more previous injuries; position, BMI and minutes played were not needed in the model (Tables 2 and 3). Because the data are observational, these are associations, not causal effects, and minutes played may itself be a consequence of injury.

The main limitations are as follows. (i) The terms were selected using the same data that were then used for the confidence intervals and p-values, which makes them somewhat optimistic. (ii) The fit is borderline on the Hosmer–Lemeshow test and the additive model ignores a clear age × training load interaction (Section 4.4), which is the natural next refinement. (iii) Club effects were not apparent, but a model with random club effects has not been fitted. (iv) The data are simulated.

Appendix: R code

A.1 Initial model (*Week_3_stats.R*)

```
# ===== DECISIONS RECORD: injury model =====
# Response: injured (1 = injured, 0 = not). It can only be yes/no.
# Model: logistic regression (binomial), because injured can only be 0 or 1
#   and a straight-line model could predict values outside 0 to 1.
# Why injury and not goals: the group left the choice open. Injury has
#   620 players (goals has only 46 matches) and several explanatory variables.
# Variables: age, position, training_load, prev_injuries, bmi, minutes_played.
#   I expect age and previous injuries to matter, e.g. players with more
#   previous injuries may be more likely to be injured again (to be tested).
# minutes_played: its coefficient was about 0 (p = 0.96), so it adds nothing
#   here. Injured players miss games, so it may be a result of injury rather
#   than a cause. I kept it in this first model and will test whether it is
#   needed before the final model.
# Diagnostics: done by my partner.
# ==========================================

# ---- Read data ----
# This line opens the file league_players and stores it as players
players <- read.csv("league_players.csv")

# ---- Set categorical variables as factors ----
# position is a label (not a number), so factor() makes R treat it as a
# category and compare each position to the baseline (DEF)
players$position <- factor(players$position)

# ---- Check the data loaded correctly ----
# dim() gives the number of rows then columns (620 players, 8 columns)
dim(players)
# str() shows every column with its type
str(players)

# ---- Injury model ----
# I am predicting injured from age, position, training load, previous
# injuries, BMI and minutes played
# family = binomial because injured can only be yes/no (1 or 0)
# The default link is logit, so coefficients are changes in log-odds
injury_model <- glm(
  injured ~ age + position + training_load +
    prev_injuries + bmi + minutes_played,
  data = players,
  family = binomial
)

# Coefficient table: estimates, standard errors, z values, p-values
summary(injury_model)
```

A.2 Model selection, final model and diagnostics (*MTH6134_GLM_analysis.R*)

```
# MTH6134 Statistical Modelling II
# Week 3: Introduction to Generalised Linear Models
# Analysis: factors associated with whether a player was injured
#
# Run this script from the folder containing league_players.csv.
# It creates a results folder containing model output and diagnostic plots.

# ============================================================
# 1. Load and inspect the players data
# ============================================================

# Read the player-level dataset. Each row represents one player.
players <- read.csv("league_players.csv")

# Treat playing position as a category, not as a numerical measurement.
players$position <- factor(players$position)

# Check the dimensions, variable types, missing values and response counts.
dim(players)
str(players)
colSums(is.na(players))
table(players$injured)

# Check that the response is coded as a binary variable (0 = no, 1 = yes).
stopifnot(all(players$injured %in% c(0, 1)))

# ============================================================
# 2. Fit the initial binomial GLM (logistic regression)
# ============================================================

# The response is injured (0/1), so a binomial GLM with the default
# logit link is appropriate. The model estimates the log-odds of injury.
# DEF is the reference position unless the factor levels are changed.
full_model <- glm(
  injured ~ age + position + training_load +
    prev_injuries + bmi + minutes_played,
  data = players,
  family = binomial(link = "logit")
)

# Display the coefficient estimates, standard errors, z statistics and p-values.
summary(full_model)

# ============================================================
# 3. Model selection
# ============================================================

# Use drop1() to test the contribution of each term to the full model
# using likelihood ratio tests. Position is tested as one overall factor.
drop1(full_model, test = "Chisq")

# Fit a simpler model retaining age, training load and previous injuries.
# These variables were retained because they showed evidence of association
# in the initial model and are substantively plausible player-level predictors.
reduced_model <- glm(
  injured ~ age + training_load + prev_injuries,
  data = players,
  family = binomial(link = "logit")
)

# Compare AIC: lower AIC indicates a better balance of fit and complexity.
AIC(full_model, reduced_model)

# Compare the nested models with a likelihood ratio test.
# A non-small p-value means the removed terms do not significantly improve fit.
anova(reduced_model, full_model, test = "Chisq")

# Use the reduced model as the final model because it has lower AIC
# and the joint likelihood ratio test does not show evidence that the
# additional terms (position, BMI and minutes played) improve the fit.
final_model <- reduced_model

# Display the final model results.
summary(final_model)

# ============================================================
```

```
# 4. Interpret coefficients as odds ratios and confidence intervals
# ============================================================

# Exponentiating logistic regression coefficients gives odds ratios.
odds_ratios <- exp(coef(final_model))

# Exponentiate the coefficient confidence intervals to obtain
# confidence intervals for the odds ratios.
or_confint <- exp(confint(final_model))

odds_ratios
or_confint

# ============================================================
# 5. Basic model fit checks
# ============================================================

# Compare residual deviance with residual degrees of freedom.
# Also calculate the Pearson chi-squared statistic divided by its df.
# These are broad checks, not definitive proof of good fit.
final_model$deviance
final_model$df.residual
sum(residuals(final_model, type = "pearson")^2)
sum(residuals(final_model, type = "pearson")^2) / final_model$df.residual

# ============================================================
# 6. Residual diagnostics
# ============================================================

# Calculate fitted probabilities and both types of GLM residuals.
fitted_prob <- fitted(final_model)
deviance_resid <- residuals(final_model, type = "deviance")
pearson_resid <- residuals(final_model, type = "pearson")
standard_deviance_resid <- rstandard(final_model, type = "deviance")

# Create a folder for output if it does not already exist.
if (!dir.exists("glm_results")) dir.create("glm_results")

# Save residual plots to a PDF. For a binary response, residuals are not
# expected to look exactly like normal linear-model residuals. Look instead
# for systematic patterns and unusually large residuals.
pdf("glm_results/residual_diagnostics.pdf", width = 8, height = 6)

plot(fitted_prob, deviance_resid,
     xlab = "Fitted probability of injury",
     ylab = "Deviance residual",
     main = "Deviance residuals versus fitted probabilities",
     pch = 19, col = "steelblue")
abline(h = 0, lty = 2)
lines(lowess(fitted_prob, deviance_resid), col = "red", lwd = 2)

plot(fitted_prob, pearson_resid,
     xlab = "Fitted probability of injury",
     ylab = "Pearson residual",
     main = "Pearson residuals versus fitted probabilities",
     pch = 19, col = "steelblue")
abline(h = 0, lty = 2)
lines(lowess(fitted_prob, pearson_resid), col = "red", lwd = 2)

# A half-normal style plot helps identify unusually large absolute
# standardised deviance residuals; it is used as a screening diagnostic.
abs_resid <- sort(abs(standard_deviance_resid))
expected <- qnorm((seq_along(abs_resid) + length(abs_resid) - 0.5) /
                    (2 * length(abs_resid) + 0.5))
plot(expected, abs_resid,
     xlab = "Expected half-normal quantiles",
     ylab = "Absolute standardised deviance residual",
     main = "Half-normal residual diagnostic",
     pch = 19, col = "steelblue")
abline(0, 1, lty = 2, col = "red")

dev.off()

# Flag observations with large standardised deviance residuals.
which(abs(standard_deviance_resid) > 2)
which(abs(standard_deviance_resid) > 3)
# ============================================================
```

```
# 7. Leverage, Cook's distance and influential observations
# ============================================================

# Hat values measure leverage: how unusual an observation's predictor
# values are relative to the rest of the data.
leverage <- hatvalues(final_model)

# Cook's distance measures how much the fitted model may change if an
# observation is omitted. It is a screening tool, not an automatic
# instruction to delete an observation.
cook <- cooks.distance(final_model)

# Common screening cut-offs:
# Cook's distance > 4/n; leverage > 2p/n, where p is the number of
# estimated model parameters (including the intercept).
n <- nrow(players)
p_parameters <- length(coef(final_model))
cook_cutoff <- 4 / n
leverage_cutoff <- 2 * p_parameters / n

# Identify observations exceeding either screening cut-off.
high_cook <- which(cook > cook_cutoff)
high_leverage <- which(leverage > leverage_cutoff)
large_residual <- which(abs(standard_deviance_resid) > 2)

# Print the cut-offs and number of observations flagged by each check.
cat("Cook's distance cut-off:", cook_cutoff, "\n")
cat("Number above Cook's distance cut-off:", length(high_cook), "\n")
cat("Leverage cut-off:", leverage_cutoff, "\n")
cat("Number above leverage cut-off:", length(high_leverage), "\n")
cat("Number with |standardised deviance residual| > 2:",
    length(large_residual), "\n")

# Combine the diagnostics in one table so potentially influential
# observations can be inspected rather than automatically removed.
diagnostic_table <- data.frame(
  row_number = seq_len(n),
  team = players$team,
  position = players$position,
  injured = players$injured,
  fitted_probability = fitted_prob,
  standard_deviance_residual = standard_deviance_resid,
  leverage = leverage,
  cooks_distance = cook
)

# Show observations flagged by at least one screening rule, ordered by Cook's D.
flagged <- diagnostic_table[
  cook > cook_cutoff |
    leverage > leverage_cutoff |
    abs(standard_deviance_resid) > 2, ]
flagged <- flagged[order(-flagged$cooks_distance), ]
print(flagged)

# Save the diagnostic table for the analysis record.
write.csv(diagnostic_table, "glm_results/observation_diagnostics.csv",
          row.names = FALSE)

# Plot leverage and Cook's distance.
pdf("glm_results/influence_diagnostics.pdf", width = 8, height = 6)

plot(leverage, type = "h",
     xlab = "Observation (row) number", ylab = "Leverage",
     main = "Leverage values", col = "steelblue")
abline(h = leverage_cutoff, col = "red", lty = 2)

plot(cook, type = "h",
     xlab = "Observation (row) number", ylab = "Cook's distance",
     main = "Cook's distance", col = "steelblue")
abline(h = cook_cutoff, col = "red", lty = 2)

plot(final_model, which = 4)  # Cook's distance against observation number
plot(final_model, which = 5)  # Residuals versus leverage with Cook's contours

dev.off()

# ============================================================
```

```
# 8. Save model output and session information
# ============================================================

# Save the model summaries and model comparison to text files.
capture.output(summary(full_model),
               file = "glm_results/full_model_summary.txt")
capture.output(summary(final_model),
               file = "glm_results/final_model_summary.txt")
capture.output(AIC(full_model, final_model),
               anova(reduced_model, full_model, test = "Chisq"),
               file = "glm_results/model_comparison.txt")

# Record the software environment to make the analysis reproducible.
capture.output(sessionInfo(),
               file = "glm_results/session_info.txt")

# End of analysis.
```

A.3 Additional checks reported in Sections 4.3 and 4.4

```
# Run after A.2 (uses players, final_model, high_cook, high_leverage).

# Hosmer-Lemeshow goodness-of-fit test with 10 groups
library(ResourceSelection)
hoslem.test(players$injured, fitted(final_model), g = 10)

# Area under the ROC curve (discrimination)
library(pROC)
auc(players$injured, fitted(final_model))

# Binned residuals (Figure 2a); the arm package uses roughly 20 bins here
library(arm)
binnedplot(fitted(final_model), residuals(final_model, type = "response"),
           nclass = 20)

# Observations flagged by both rules, and outcomes of Cook's flagged players
sum(high_cook %in% high_leverage)
table(players$injured[high_cook])

# Club effect: does adding team improve the final model?
players$team <- factor(players$team)
anova(final_model, update(final_model, . ~ . + team), test = "Chisq")

# Linearity: quadratic terms and training load as a factor
age_c <- players$age - mean(players$age)
anova(final_model, update(final_model, . ~ . + I(age_c^2)), test = "Chisq")
anova(final_model, update(final_model, . ~ . + I(prev_injuries^2)),
      test = "Chisq")
anova(final_model, update(final_model, . ~ . + I((training_load - 3)^2)),
      test = "Chisq")
anova(final_model,
      update(final_model, . ~ . - training_load + factor(training_load)),
      test = "Chisq")

# Pairwise interactions among the retained predictors
anova(final_model, update(final_model, . ~ . + age:training_load),
      test = "Chisq")
anova(final_model, update(final_model, . ~ . + age:prev_injuries),
      test = "Chisq")
anova(final_model, update(final_model, . ~ . + training_load:prev_injuries),
      test = "Chisq")

# Interaction model: coefficients and AIC comparison
interaction_model <- update(final_model, . ~ . + age:training_load)
summary(interaction_model)
AIC(final_model, interaction_model)

# Odds ratio for training load at a given age under the interaction model
b <- coef(interaction_model)
exp(b["training_load"] + b["age:training_load"] * c(22, 25, 30))
```